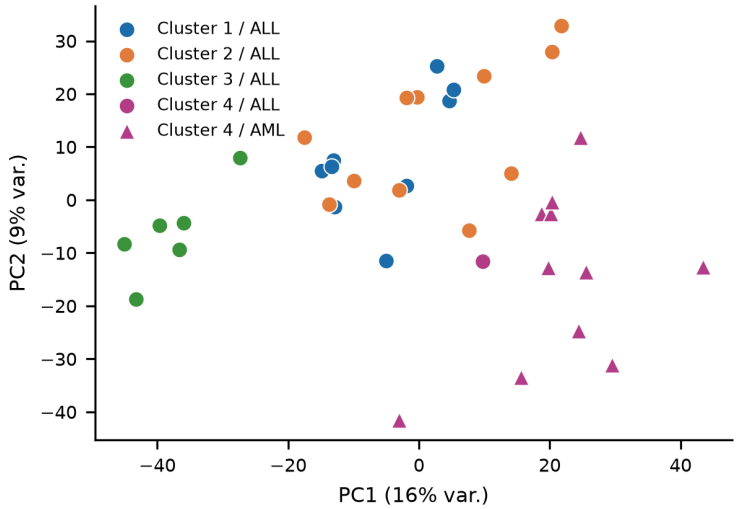
Clustering the Golub Leukemia Samples

$n = 38$ samples, $p = 3{,}051$ genes, no missing values · standardized genes, PCA (21 PCs, 80.6% var.) · seed 432

Methods. Samples were already scaled but genes were not, so each gene was standardized; because $p \gg n$, clustering used the first 21 PCs. Three algorithms were compared: K-means (50 restarts), hierarchical clustering (average linkage, Euclidean distance), and spectral clustering (k-nearest-neighbor affinity, $n_{\text{neighbors}} \in \{5, 10, 15\}$), each for $K = 2, \ldots, 7$. Solutions with a cluster smaller than 5% of n were rejected. Selection: highest mean silhouette; ties broken by stability (mean ARI over 20 seeds), then smaller K. Labels were not used until after selection.

Selection evidence. All six hierarchical solutions isolated a single sample and were rejected. The best valid setting for each remaining method:

Method	Setting	Mean silhouette	Stability (ARI)
K-means	$K = 4$	**0.121**	**0.70**
Spectral (kNN)	$K = 5$, $n_{\text{neighbors}} = 5$	0.114	1.00



(a) K-means ($K = 4$) clusters on the first two PCs. Color = cluster; shape = leukemia type (added after selection).

Table 1: Clusters vs. known type (ARI = 0.33)

Cluster	ALL	AML
1	9	0
2	11	0
3	6	0
4	1	11

Interpretation. Cluster 4 is the AML group: it contains all 11 AML samples and only 1 of the 27 ALL samples, and it lies at high PC1 and low PC2. Clusters 1–3 split the ALL samples into three groups. Cluster 3 (6 samples) is clearly separated at low PC1. Clusters 1 and 2 overlap in this view, so the differences between them lie in later components. The ARI of 0.33 is low mainly because the ALL samples are split into three clusters; the AML/ALL boundary itself is almost perfectly recovered.

Limitations.

- All silhouettes are low (≤ 0.12), so the cluster structure is weak; K-means and spectral clustering scored very close.
- K-means at $K = 4$ is only moderately stable across seeds (ARI 0.70), and results may change with a different variance cutoff or K.
- With only 38 samples, small clusters (e.g., cluster 3) rest on few observations, and no available label can confirm the three ALL subgroups.

Appendix: complete skill definition (`SKILL.md`)

```
---
name: clustering-report
description: Compare clustering algorithms on an unlabeled dataset, select one, and write a
 ↪  one-page report.
disable-model-invocation: true
---

# Clustering Analysis Report
Run only when the user invokes this skill. The user supplies: dataset path,
feature columns, ID column, optional label column (for interpretation only),
seed, output folder. Ask if any are missing. Never assume dataset-specific facts; base every
 ↪  statement on computed results.

## Tools
Python (pandas, numpy, scikit-learn, matplotlib). One script must reproduce every number,
 ↪  figure, and table.

## Analysis rules
1. **Load and check.** Report n, p, missing values. Fit and select using
   features only; never use labels until the final interpretation step.
2. **Preprocessing.** standardize features unless it is already standardized. When p >> n,
 ↪  consider variance filtering and/or PCA. Justify the choice.
3. **Candidate algorithms.** At least three different clustering algorithms (PCA/UMAP alone does
 ↪  not count). K-means, hierarchical (linkage: average; distance: Euclidean), spectral
 ↪  (k-nearest-neighbors).
4. **Tuning and number of clusters.** Try K = 2..min(10, n/5) and spectral n_neighbors in {5,
 ↪  10, 15}; choose K and n_neighbors by mean silhouette.
5. **Compare and select.** Select the highest mean silhouette; break ties with stability (ARI
 ↪  across 20 seeds), then the smaller K. Reject solutions with any cluster smaller than 5% of
 ↪  n.
6. **Interpret.** Only for the selected method: cluster sizes, what separates
   the clusters, and (if labels exist) a cross-tab of clusters vs labels.

## Report
Limit the report to **one printed page**, including all text, figures,
and tables. Show clustering results from **only one selected method**;
other methods may be named briefly to justify the choice, with no
results shown for them. format: Quarto `.qmd` -> PDF; letter size,
11pt, 0.75 in margins; hide code. Render and confirm one page; if too long,
cut text first. Sections, in order:
1. **Title** + one line: n, p, preprocessing, seed.
2. **Methods** (2-4 sentences): candidates, tuning, selection criteria.
3. **Selection evidence** small table (method, K,
   criterion score). Bold the selected method.
4. **Figure** (selected method only, readable). PC1 vs PC2
   colored by cluster (label as marker shape).
5. **Interpretation** (2-3 sentences): what the groups are; cross-tab vs labels.
6. **Limitations** (2-4 bullets). small n, sensitivity to
   preprocessing/K, labels used only after selection.
```